Experimental Report: Transfer90 Semantic Watermark Inheritance

Updated:27 October2026. **Scope:**seed-0 SD2.1 teacher to ordinary SD1.4 LoRA student. Transfer90 completed at 10: 33 ICT. This report follows the experimental structure of the supplied October11 English report; its SERUM/CtrlRegen numbers belong to a different experiment and are not pooled here.

1. Main findings

Transfer90 reaches97.8% single-image detection (978/1000), Wilson95%CI[96.69,98.54]%. Both the estimate and its lower confidence bound exceed90%. The score threshold,8-bit key and semantic verifier are unchanged from the comparable k=1 campaign. This is the highest detection among the comparable completed branches listed here.

Against Structure 12500:detection increases6.7 percentage points, FID improves52.89584 to52.56332, and CLIP increases0.786258 to0.786725. LPIPS worsens0.528817 to0.539145; KID is effectively unchanged at the displayed scale. Transfer90 therefore does not improve every quality metric.

Against the matched-panel k=1 baseline:detection increases22.8 percentage points, LPIPS falls0.550447 to0.539145, and FID/KID improve. CLIP is slightly lower. Longer training, a different teacher and a different LR schedule change together; the result is a system-level comparison, not an isolated effect of the three-view loss.

2. Experimental protocol

Component	Completed setting
Teacher	Manojb/stable-diffusion-2-1-base; warm-start completed Projected native(k=1), then 800 additional refinement updates; teacher LoRA rank32.
Student	Pinned CompVis/stable-diffusion-v1-4, newly initialized LoRA rank32 attention+convolution; frozen VAE/text encoder; ordinary denoising MSE.
Data	5000 teacher training images; 1000 validation; 1000 test; 5000 clean calibration prompts. Same test prompt panel as Structure 12500 and k=1baseline.
Teacher optimization	AdamW LR5e-6, batch4; auxiliary native objective every2 updates; microbatch1; full 50-step PNDM forward, derivatives through last 10 denoiser calls.
Student optimization	15000 updates, batch4, fp16, seed0; LR1e-4through 10000, cosine to2e-5by 15000.
Selection / generation	Minimum validation denoising MSE at 7500/10000/12500/15000; selected 15000. PNDM50, CFG7.5,512x512, batch16; student seed500000+index.

3. Method: native preservation and transfer-aware teacher refinement

Lineage. The original k=1 teacher checkpoint3000 was refined for 600 updates to obtain Projected native. Transfer90 warm-starts that adapter and adds 800 updates. Structure 12500 uses a separate Structure/margin teacher refined 600 updates directly from the original k=1 checkpoint. Transfer90 is not a continuation of the Structure 12500 student.

Inherited watermark objective. The original threshold_cfg_v1 loss preserves clean conditional/unconditional teacher predictions and applies calibrated semantic shifts to predicted clean latents and their teacher-VAE cycle. The semantic term operates on timesteps up to500. The downstream student receives only images/captions and ordinary denoising supervision.

Native image preservation. For each training prompt/seed, generate the full PNDM trajectory and compare the final RGB image against its clean SD2.1 paired image. RGB quantization is represented differentiably. Only the last 10 denoiser calls carry gradients; the forward pass and PNDM history remain complete. This is inspired by DRaFT-K[1], not a shortened 10-step sampler.

```
L_quality = LPIPS(x_marked,x_clean) +0.1*MSE_RGB(x_marked,x_clean)
+0.25*(1-MS_SSIM(x_marked,x_clean))
```

Three teacher-side views.(1)Native marked RGB;(2)stochastic SD1.4 VAE encode/decode;(3)one noising/denoising reconstruction through a frozen historical SD1.4 proxy, cycling timesteps 50,150,300 and using CFG7.5. The proxy is the Projected native 10000 student trained on historical watermarked images. Its weights are frozen; gradients pass through it to the teacher image. The final student is separately initialized and is never used as this proxy.

```
score_v = mean_j(key_j * standardized_coordinate_vj)
d_v = max(target_v-score_v,0)/sigma_clean
L_margin =0.5*mean_v(d_v^2) +0.5*max_v(d_v^2)
L_bits = mean_v,j[max(0,0.25-key_j*coordinate_vj)^2]
L_owner = L_margin +0.1*L_bits
```

Score floors. Let tau=max(frozen teacher threshold, frozen student threshold) and r=min(1,(update+1)/200). Native/cycle floor=tau+(1+r)*sigma_clean; denoised floor=tau+(0.5+0.5r)*sigma_clean. The native target also preserves the previous Projected native image score when that score exceeds the floor. Deficits are penalized per image before averaging; the worst of the three views receives extra weight. This finite-view objective is not a population CVaR guarantee.

Gradient balancing and projection. Compute base, quality and ownership gradients separately. Scale auxiliary gradients to target1.0and1.5 times the base-gradient norm, with coefficient cap100. AdamW proposes an update; on auxiliary steps, project its actual parameter displacement onto the intersection of the three local halfspaces g_i dot delta<=0. This is inspired by gradient surgery[2], but operates on the realized AdamW displacement. It provides local truncated-gradient constraints, not a monotonic detection/quality guarantee.

4. Final student results and paired comparisons

Branch	Steps	Detected	TPR /95%CI	LPIPS	FID	KID x 1000	CLIP
Baseline k=1	7500	750/1000	75.0%	0.550447	54.444	1.6665	0.787681
Structure 12500	12500	911/1000	91.1% /89.17-92.71	0.528817	52.896	0.6842	0.786258
Transfer90	15000	978/1000	97.8% /96.69-98.54	0.539145	52.563	0.6847	0.786725

Metric definitions.LPIPS(AlexNet0.1)[3]compares each student image with unadapted SD1.4 at the same prompt/seed; lower means closer, not necessarily more attractive. FID[4]/KID[5]compare 1000 student outputs with clean SD2.1 teacher outputs, not real COCO. CLIPScore=2.5*max(cosine(image, text),0), CLIPViT-B/32. Teacher LPIPS uses a different reference(clean SD2.1), so its magnitude must not be ranked against student LPIPS.

Transfer90 minus	TPR delta pp /95%CI	LPIPS delta /95%CI	Gained/lost detection	Images LPIPS improved
Baseline k=1	+22.8 [+20.0,+25.6]	-0.01130 [-0.01666,-0.00595]	240/12	56.7%
Structure 12500	+6.7 [+4.9,+8.6]	+0.01033 [+0.00471,+0.01601]	78/11	45.9%
Structure 7500	+10.4 [+8.4,+12.4]	+0.02258 [+0.01706,+0.02804]	112/8	35.0%
Projected native 7500	+18.6 [+16.1,+21.1]	+0.01693 [+0.01151,+0.02249]	193/7	39.0%

Paired percentile bootstrap: 5000 prompt-index resamples, seed20261027. Generation settings and clean-reference hashes match for the main three-way comparison. These intervals describe variation across sampled prompts for fixed trained models; they do not measure retraining-seed variation or correct adaptive test reuse.

5. Frozen detector, clean controls and bit recovery

Student threshold tau=0.8541307449340821, calibration target FPR1% on 5000 clean prompts. The empirical held-out FPR is0.8%(8/1000)for unadapted SD1.4 and0.2%(2/1000)for the historical clean-trained 7500 student. Their Wilson95%CIs are[0.406,1.571]% and[0.055,0.726]%. A matched clean-trained15000control is missing.

Transfer90 per-image bit accuracy is91.9125%; exactly all8bits are recovered in51.6%of images. Signs of the mean coordinates recover8/8bits(100%aggregate key accuracy). **These three quantities differ from97.8%single-image threshold detection.** A100%aggregate bit figure must not be presented as100% image-level detection.

6. Comparison with completed historical branches

Comparable-panel k=1 experiments: 1000 test images, the same student score threshold and metric reference conventions. Budgets and teacher recipes differ; rows are descriptive operating points. No attribution to one component follows from ranking this table.

Student branch	Updates	TPR%	LPIPS	FID	KID x 1000	CLIP
Baseline k=1	7500	75.0	0.550447	54.444	1.6665	0.787681
Quality	7500	80.8	0.546656	55.281	1.7606	0.785113
Continuation	7500	82.3	0.550218	54.887	1.4148	0.786429
Native full	7500	80.5	0.544096	54.605	1.2649	0.786889
Projected native 7500	7500	79.2	0.522219	53.358	0.7950	0.785335
Projected native 10000	10000	82.1	0.533119	53.453	0.7732	0.787516
Structure 7500	7500	87.4	0.516560	52.735	0.5454	0.788896
Structure 10000 warm restart	10000	86.4	0.547930	53.851	0.9707	0.785565
Structure 12500	12500	91.1	0.528817	52.896	0.6842	0.786258
scale085_7500	7500	42.7	0.533659	54.563	1.4637	0.787436
scale085_long22500	22500	52.3	0.573231	54.858	2.0206	0.790023
scale085_conv_restore	7500	57.5	0.535404	54.230	1.3024	0.788586
quality_gradient	7500	80.7	0.543704	55.377	1.6724	0.786408
quality_cka	7500	80.5	0.548790	54.939	1.6421	0.784089
quality_transfer	7500	80.5	0.545900	55.087	1.4964	0.786084
Transfer90	15000	97.8	0.539145	52.563	0.6847	0.786725

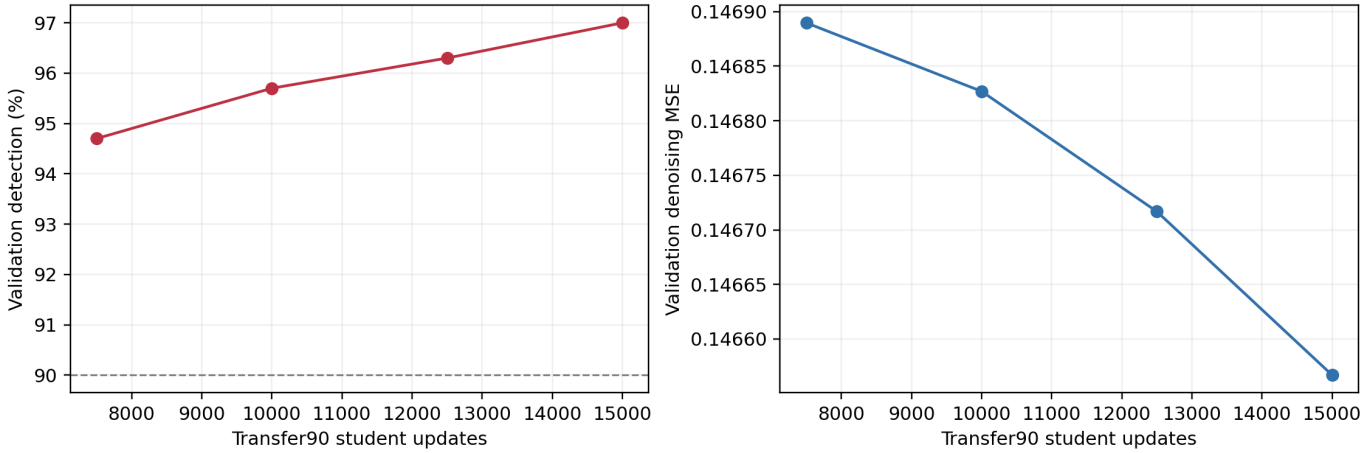
Earlier99.70%result. The historical lora_r32_conv_wm row in result_vast used a selected kappa=2 teacher and a different artifact/panel configuration: TPR99.70%, FID62.676, KID0.0069348, CLIP0.79334. It is preserved as historical context, not the matched k=1baseline. Student LPIPS is not supplied in that row. Therefore we cannot claim Transfer90 improves its LPIPS or reproduces its setup.

Lessons supported by the runs. Uniform scale0.85reduced detection sharply; longer student training did not consistently repair it. Native image objectives improved paired fidelity more reliably than the earlier proxy-only refinements. Structure 7500 retains better LPIPS than Transfer90; Transfer90 prioritizes inheritance and reaches a much stronger detection operating point. Neither outcome proves that all visual artifacts have been removed.

7. Teacher evidence and student learning curve

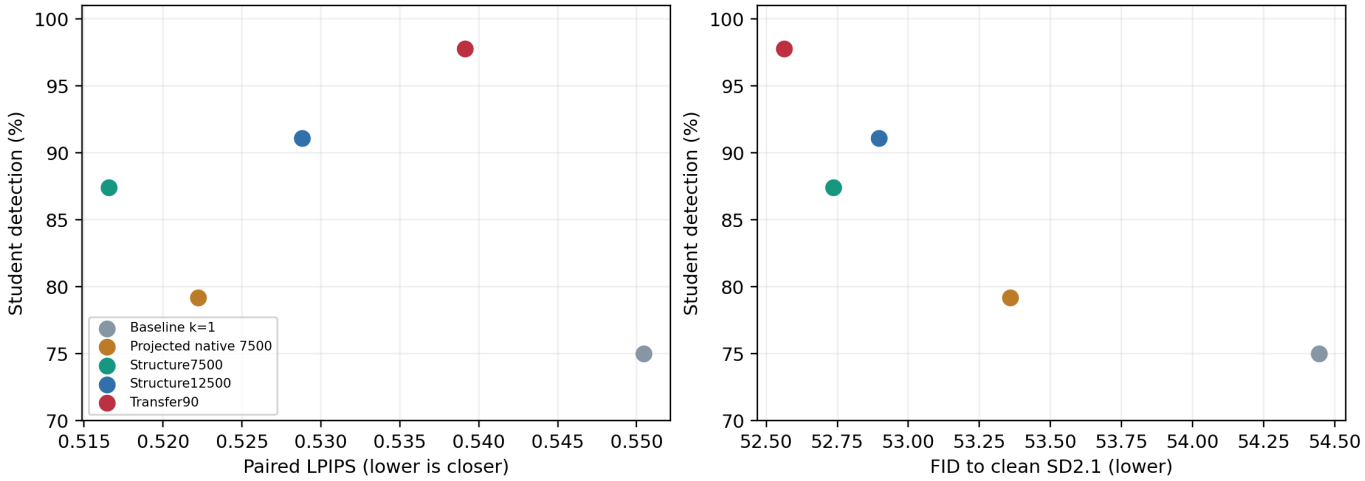
Measure	Transfer90 teacher
Validation detection	998/1000 (99.8%)
Test detection	999/1000 (99.9%); teacher threshold0.8509314060
Validation LPIPS / CLIP	0.367454 / 0.781779; clean teacher CLIP 0.782060
Interpretation	Teacher detection is strong, but is not a substitute for downstream student detection; teacher FID/KID not included in this exported completed summary.

Student checkpoint	Validation MSE	Validation TPR	Selected
7500	0.146889691	94.7%	No
10000	0.146826940	95.7%	No
12500	0.146716820	96.3%	No
15000	0.146566953	97.0%	Yes



At 12500 updates, Transfer90 validation detection is96.3%, versus91.3% for Structure 12500 on validation. This reduces the concern that the entire difference appears only after the extra2500 updates, but remains a teacher-and-training-recipe comparison. A fixed 12500 Transfer90 test endpoint was not evaluated by this report; do not substitute validation96.3% for its test TPR.

8. Quality/detection trade-offs and limitations



Transfer90 is strongest in detection among the displayed comparable branches; Structure 7500 lies farther left in LPIPS. FID values are close at this 1000-image sample size. Small KID/FID gaps are point estimates, not evidence of a statistically established ranking. KID subset standard deviation is not a confidence interval across training seeds.

Visual audit scope. The following20pages show the first 20 test prompts with no selection by attractiveness, score or detection. Each page includes clean/marked SD2.1teachers and unadapted SD1.4, then baseline k=1, Structure 12500 and Transfer90students. Teacher seeds use image IDs; student seeds are500000+index. Pairing holds within teacher and student families; teacher-vs-student noise is not identical. The gallery retains all120original512x512PNGs; PDF images use high-quality JPEG previews for portability.

Limits. One training seed and repeatedly inspected test prompts; no fresh confirmatory holdout. Teacher source, refinement duration, student budget and LR schedule change together. The frozen proxy is a surrogate rather than differentiated full student training. No matched 15000 clean control, robustness-to-distortion sweep or systematic human preference/anatomy evaluation is completed. Per-image LPIPS is reference distance; altered composition can raise it even if an image is plausible.

Supported statement. Under the current seed-0 protocol, Transfer90 preserves strong image-level semantic detection through ordinary LoRA adaptation while improving LPIPS/FID/KID relative to the matched k=1baseline. It raises detection above Structure12500at the cost of worse paired LPIPS. The evidence supports a stronger inheritance operating point, not a universal image-quality improvement or a reproduction of the historical kappa=2result.

9. Evidence, reproducibility and next checks

The evidence bundle includes completed student/teacher JSONs, verifier report, baseline summary, source code, all generation manifests,20 matched prompts and SHA256checks for120 images. Student adapter SHA256: 960564f91f2d132a139ff5beb7c3b202f8955fe18f0f488d57dc3473c8b7adb9. The snapshot was selected only by validation MSE0.1465669530. Current server location:/var/lib/wmq/logs/project_native_transfer90/runs/run_20261026_114758_661241/transfer90. Archived manifests retain the old/root/logs paths; relocation requires path repair before rerunning.

Priority validation: repeat seeds and a fresh holdout; add matched 15000 clean adaptation; compare fixed budgets and leave-one-component-out teacher ablations; perform blinded human comparisons of the same prompt/seed panel. Checkpoint averaging was cancelled and is not part of these results.

Primary methodological references:
[1]Clark et al., Directly Fine-Tuning Diffusion Models on Differentiable Rewards(ICLR2024), arxiv.org/abs/2309.17400.
[2]Yu et al., Gradient Surgery for Multi-Task Learning(NeurIPS2020), arxiv.org/abs/2001.06782.
[3]Zhang et al., The Unreasonable Effectiveness of Deep Features as a Perceptual Metric(CVPR2018), arxiv.org/abs/1801.03924.
[4]Heusel et al., GANs Trained by a Two Time-Scale Update Rule...(NeurIPS2017), arxiv.org/abs/1706.08500.
[5]Binkowski et al., Demystifying MMD GANs(ICLR2018), arxiv.org/abs/1801.01401.
Channel motivation: HMARK(arxiv.org/abs/2512.00094)and Radioactive Watermarks in Diffusion and Autoregressive Image Generative Models(arxiv.org/abs/2506.23731). Their reported metrics are not directly comparable to this experiment.

10.1. Paired visual comparison: sample00

Prompt:A battered refrigerator stands open in a dirty room.

Teacher seed(image ID): 45663; student seed: 500000. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.3810 | detect



Unadapted SD1.4
Score-0.4538 | below threshold



Baseline k=1 student
Score1.3886 | detect



Structure12500 student
Score1.4780 | detect



Transfer90 student
Score1.5847 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.2. Paired visual comparison: sample01

Prompt:a person cutting a pink cake with pink roses

Teacher seed(image ID): 209548; student seed: 500001. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.5391 | detect



Unadapted SD1.4
Score-0.3453 | below threshold



Baseline k=1 student
Score1.0530 | detect



Structure12500 student
Score1.4025 | detect



Transfer90 student
Score1.6534 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.3. Paired visual comparison: sample02

Prompt:A man in a green shirt stands by a girl holding a piece of cake on a plate.

Teacher seed(image ID): 313034; student seed: 500002. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.5572 | detect



Unadapted SD1.4
Score0.2185 | below threshold



Baseline k=1 student
Score0.9931 | detect



Structure12500 student
Score1.0473 | detect



Transfer90 student
Score1.7213 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.4. Paired visual comparison: sample03

Prompt: Two children under a " Sesame Street" sign on a light post.

Teacher seed(image ID): 217137; student seed: 500003. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score2.0137 | detect



Unadapted SD1.4
Score-0.3142 | below threshold



Baseline k=1 student
Score0.8261 | below threshold



Structure12500 student
Score1.2586 | detect



Transfer90 student
Score1.4583 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.5. Paired visual comparison: sample04

Prompt: The red plate holds rice, tofu, broccoli and water chestnuts.

Teacher seed(image ID): 205435; student seed: 500004. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score2.0868 | detect



Unadapted SD1.4
Score-0.4499 | below threshold



Baseline k=1 student
Score1.0630 | detect



Structure12500 student
Score1.6624 | detect



Transfer90 student
Score1.5824 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.6. Paired visual comparison: sample05

Prompt:A man holding a Nintendo Wii controller next to a pretty young girl holding another Nintendo Wii controller.

Teacher seed(image ID): 335785; student seed: 500005. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.4950 | detect



Unadapted SD1.4
Score-0.4139 | below threshold



Baseline k=1 student
Score1.2085 | detect



Structure12500 student
Score1.3925 | detect



Transfer90 student
Score1.9255 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.7. Paired visual comparison: sample06

Prompt:A sad, young girl sits on her bed, moping.

Teacher seed(image ID): 118542; student seed: 500006. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.9818 | detect



Unadapted SD1.4
Score-0.6296 | below threshold



Baseline k=1 student
Score0.8474 | below threshold



Structure12500 student
Score1.3937 | detect



Transfer90 student
Score1.5480 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.8. Paired visual comparison: sample07

Prompt: Two tall teddy bears standing next to each other with a person kneeling between them.

Teacher seed(image ID): 227801; student seed: 500007. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score2.0997 | detect



Unadapted SD1.4
Score0.3178 | below threshold



Baseline k=1 student
Score0.3586 | below threshold



Structure12500 student
Score0.7873 | below threshold



Transfer90 student
Score1.7819 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.9. Paired visual comparison: sample08

Prompt:a couple of cows stand on some grass

Teacher seed(image ID): 576564; student seed: 500008. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.6034 | detect



Unadapted SD1.4
Score0.3855 | below threshold



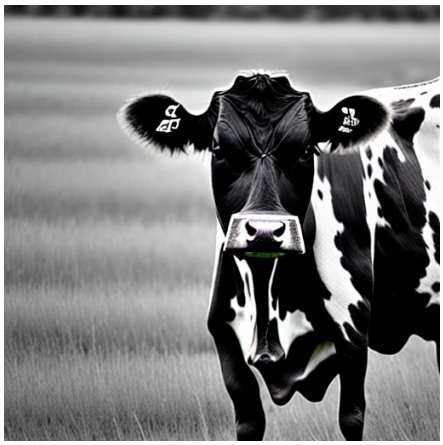
Baseline k=1 student
Score0.6821 | below threshold



Structure12500 student
Score0.2035 | below threshold



Transfer90 student
Score0.8872 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.10. Paired visual comparison: sample09

Prompt: Several stuffed animals handing from a metal rail.

Teacher seed(image ID): 208844; student seed: 500009. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.9535 | detect



Unadapted SD1.4
Score-0.0563 | below threshold



Baseline k=1 student
Score0.8035 | below threshold



Structure12500 student
Score1.0068 | detect



Transfer90 student
Score1.7902 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.11. Paired visual comparison: sample10

Prompt:A baby girl standing in a shopping cart holding an umbrella.

Teacher seed(image ID): 443969; student seed: 500010. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.8741 | detect



Unadapted SD1.4
Score0.3979 | below threshold



Baseline k=1 student
Score0.9064 | detect



Structure12500 student
Score1.4001 | detect



Transfer90 student
Score1.7779 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.12. Paired visual comparison: sample11

Prompt:A bus travels on a road near buildings and parked vehicles

Teacher seed(image ID): 334405; student seed: 500011. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score2.0515 | detect



Unadapted SD1.4
Score-0.0800 | below threshold



Baseline k=1 student
Score0.9908 | detect



Structure12500 student
Score1.1372 | detect



Transfer90 student
Score1.7329 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.13. Paired visual comparison: sample12

Prompt:a small child is standing on a skateboard

Teacher seed(image ID): 507935; student seed: 500012. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.8140 | detect



Unadapted SD1.4
Score-0.1173 | below threshold



Baseline k=1 student
Score1.2769 | detect



Structure12500 student
Score1.5243 | detect



Transfer90 student
Score1.5088 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.14. Paired visual comparison: sample13

Prompt:A dog is standing in floodwater beside a house.

Teacher seed(image ID): 184606; student seed: 500013. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.6828 | detect



Unadapted SD1.4
Score-0.3038 | below threshold



Baseline k=1 student
Score0.9495 | detect



Structure12500 student
Score1.2599 | detect



Transfer90 student
Score1.2114 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.15. Paired visual comparison: sample14

Prompt:A room filled with furniture and hardwood floors.

Teacher seed(image ID): 15947; student seed: 500014. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.9308 | detect



Unadapted SD1.4
Score0.1993 | below threshold



Baseline k=1 student
Score0.9989 | detect



Structure12500 student
Score1.3411 | detect



Transfer90 student
Score1.8747 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.16. Paired visual comparison: sample15

Prompt:A street containing various cars and lampposts at night.

Teacher seed(image ID): 372316; student seed: 500015. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.6491 | detect



Unadapted SD1.4
Score-0.6901 | below threshold



Baseline k=1 student
Score1.0850 | detect



Structure12500 student
Score1.1896 | detect



Transfer90 student
Score1.1527 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.17. Paired visual comparison: sample16

Prompt: Two elephants are near a watering hole at the same time

Teacher seed(image ID): 309883; student seed: 500016. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.6564 | detect



Unadapted SD1.4
Score0.0534 | below threshold



Baseline k=1 student
Score1.0963 | detect



Structure12500 student
Score1.7078 | detect



Transfer90 student
Score1.8760 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.18. Paired visual comparison: sample17

Prompt:A wooden bench and a fence with some flowers.

Teacher seed(image ID): 540782; student seed: 500017. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.6689 | detect



Unadapted SD1.4
Score0.1000 | below threshold



Baseline k=1 student
Score1.4817 | detect



Structure12500 student
Score1.0209 | detect



Transfer90 student
Score1.6787 | detect



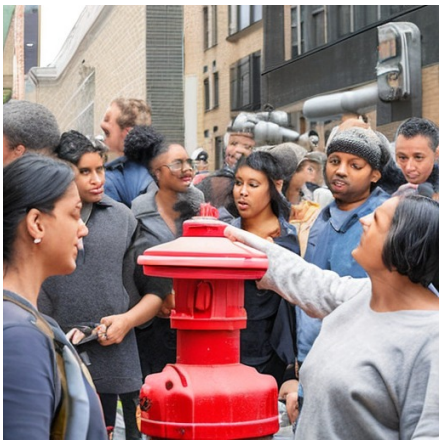
Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.19. Paired visual comparison: sample18

Prompt:A crowd of people standing near a fire hydrant.

Teacher seed(image ID): 551315; student seed: 500018. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

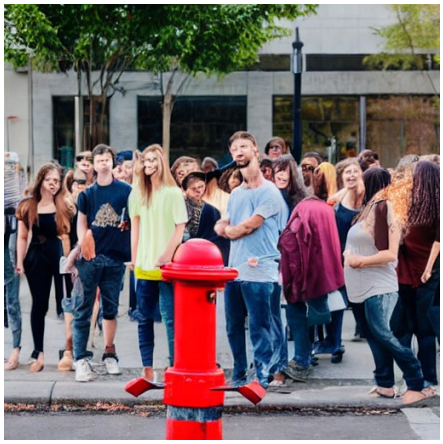
Clean SD2.1 teacher



Transfer90 teacher
Score1.8846 | detect



Unadapted SD1.4
Score-0.2289 | below threshold



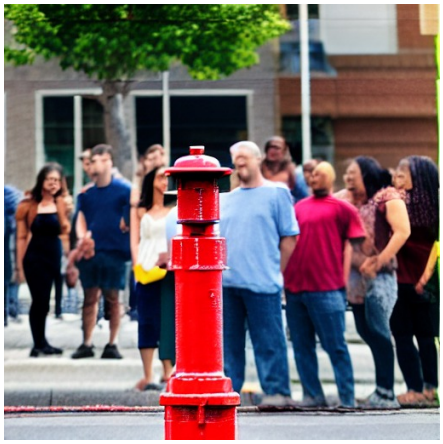
Baseline k=1 student
Score1.1211 | detect



Structure12500 student
Score1.2570 | detect



Transfer90 student
Score1.5437 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.

10.20. Paired visual comparison: sample19

Prompt:A toddler with a fork in her mouth and a picnic table and people behind her.

Teacher seed(image ID): 1667; student seed: 500019. Teacher-vs-student seeds differ. First 20 test samples; no visual filtering.

Clean SD2.1 teacher



Transfer90 teacher
Score1.6171 | detect



Unadapted SD1.4
Score-0.1652 | below threshold



Baseline k=1 student
Score0.8381 | below threshold



Structure12500 student
Score1.5419 | detect



Transfer90 student
Score1.2784 | detect



Score measures watermark evidence, not image quality. Inspect composition, faces, hands, textures and prompt fidelity in the paired images. Full-resolution original PNGs are available in the accompanying gallery.